Example 3.47 Varshika drew 6 circles with different sizes. Draw a graph for the relationship between the diameter and circumference (approximately related) of each circle as shown in the table and use it to find the circumference of a circle when its diameter is 6 cm.

Diameter (x) cm	1	2	3	4	5
Circumference (y) cm	3.1	6.2	9.3	12.4	15.5

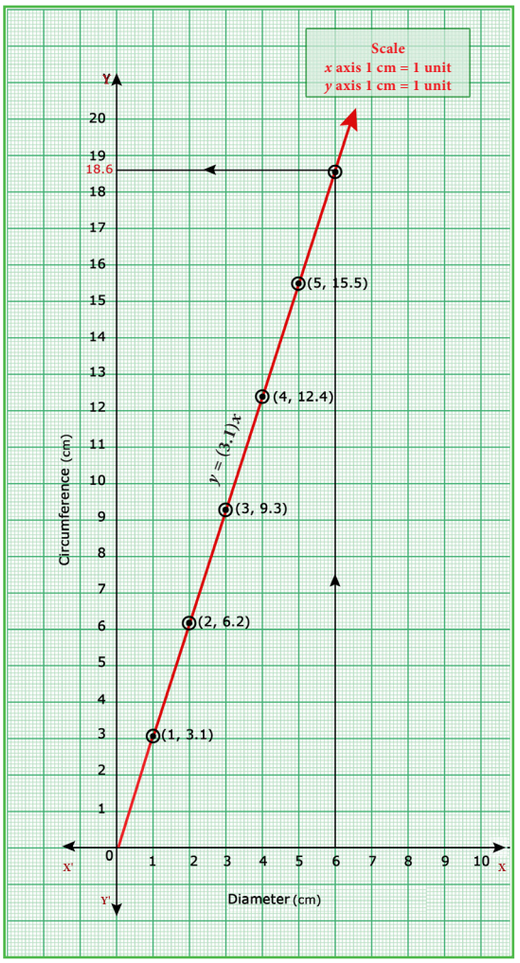


Fig. 3.10

Solution:

From the table, we found that as x increses, y also increases. Thus, the variation is a **direct variation.**

Let $y = kx$, where k is a constant of proportionality.

From the given values, we have,

$$k = \frac{3.1}{1} = \frac{6.2}{2} = \frac{9.3}{3} = \frac{12.4}{4} = \ldots = 3.1$$

When you plot the points (1, 3.1) (2, 6.2) (3, 9.3), (4, 12.4), (5, 15.5), you find the relation $y = (3.1)x$ forms a straight-line graph.

Clearly, from the graph, when diameter is 6 cm, its circumference is 18.6 cm.

Example 3.48 A bus is travelling at a uniform speed of 50 km/hr. Draw the time-distance graph and hence find

(i) the constant of variation
(ii) how far will it travel in 90 minutes?
(iii) the time required to cover a distance of 300 km from the graph.

Solution

Let x be the time taken in minutes and y be the distance travelled in km.

Time taken x (in minutes)	60	120	180	240
Distance y (in km)	50	100	150	200

(i) Observe that as time increases, the distance travelled also increases. Therefore, the variation is a direct variation. It is of the form $y = kx$.

Constant of variation

$$k = \frac{y}{x} = \frac{50}{60} = \frac{100}{120} = \frac{150}{180} = \frac{200}{240} = \frac{5}{6}$$

10th_Maths_Chapter 3_English.indd 125 12/23/2024 5:33:53 PM